

Q.21 Discuss dendrogram and elbow method in brief.
(CO4)

Q.22 What are the components of time-series data? (CO4)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 What do you understand by learning systems?
Explain process of building learning systems in detail. (CO1)

Q.24 Explain the difference between K-mean clustering and Hierarchical Clustering. (CO3)

Q.25 What is forecasting? Discuss various forecasting techniques in detail. (CO4)

No. of Printed Pages : 4

223842

Roll No.

4th Sem / Artificial Intelligence & Machine Learning

Subject : Machine Learning

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Application of machine learning methods to large databases is called: (CO1)

- a) Data mining.
- b) Big data computing
- c) artificial intelligence
- d) Internet of things

Q.2 In what type of learning labelled training data is used? (CO1)

- a) unsupervised learning
- b) reinforcement learning
- c) Supervised learning
- d) active learning

Q.3 Which of the following is the correct extension of the Python file? (CO2)

- a) .python
- b) .pl
- c) .py
- d) .p

- Q.4 Which of the following is a goal of clustering algorithms? (CO3)
- Classification
 - Regression
 - Dimensionality reduction
 - Grouping similar data points together
- Q.5 In which category does linear regression belong to? (CO4)
- Neither supervised nor unsupervised learning
 - Both supervised and unsupervised learning
 - Unsupervised learning
 - Supervised learning
- Q.6 What does the term "univariate time series" refer to in time series forecasting? (CO3)
- multivariate time series
 - time series visualization
 - time series modeling
 - single variable time series

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define mean, median and mode. (CO2)
- Q.8 What are tuples in python? (CO2)

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- Q.9 Which clustering algorithm is capable of detecting outliers as noise points? (CO3)
- Q.10 Give full-form of AR-ARMA. (CO4)
- Q.11 What are sets and dictionaries? (CO2)
- Q.12 What is a dataset in python? (CO2)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 What is the difference between supervised and unsupervised learning? (CO1)
- Q.14 What is Python? List some popular applications of Python. (CO2)
- Q.15 How comments are made in Python and what is its use? (CO2)
- Q.16 Explain some uses of machine learning. (CO1)
- Q.17 Explain various conditional statements used in python with suitable example. (CO2)
- Q.18 Explain Map and Filter function in functional programming. (CO3)
- Q.19 Explain the concepts of handling missing values in datasets. (CO3)
- Q.20 Explain bar graphs and histogram in data visualization. (CO4)

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